Model Predictive Control: Physics-Informed Surrogate for Photovoltaic Panel Orientation

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Abstract

We build a model predictive controller (MPC) for a 3-axis PV mount, using a neural surrogate of a lumped thermal model. The surrogate matches its generator to **0.06 K** at 15 min. The power-optimal pose is always the max-irradiance pose, so the surrogate's real job is a 15-min temperature forecast. On a year of Sandia data, a physics-calibrated model beats persistence (**2.21 vs 3.20 K** RMSE); the zero-shot network does not. In a closed-loop sandbox the MPC captures **98.7%** of oracle energy.

Introduction

MPC uses a model of the system to forecast its behavior over a finite *prediction horizon*, then picks the best action (Nghiem et al., 2023). Our loop runs every 15 minutes:

1. **Measure the state (weather, panel temperature, pose).**
2. **Predict the next 15 min for each candidate pose.**
3. **Choose the pose that maximizes electrical power.**
4. **Apply only the first move, then repeat.**

Re-planning every step keeps forecast error from accumulating, like chess: make one move, then re-read the board.

Model

Lumped energy balance (per unit area):

$$C\frac{dT}{dt} = \alpha G - \eta(T)\,G - h(v)\,(T - T_{amb})$$

$$\eta(T) = \eta_{ref}\left[1 - \beta\,(T - T_{STC})\right]$$

$$h(v) = h_0 + k\,(v - 1\,\mathrm{m/s})$$

η falls with T (Sze & Lee, 2012; Yadav et al., 2023); h_0 from NOCT, k from McAdams.

Irradiance from I_{SC} (Laudani et al., 2021):

$$G = G_{STC}\,\frac{I_{SC}}{I_{SC,STC}\left[1 + \alpha_I (T - T_{STC})\right]}$$

Surrogates and MPC Objective

MLP (8 inputs → T_{ss}, T_{15}), trained by MSE on 10^6 RK4 rows:

$$\mathscr{L}_{MLP} = \frac{1}{N}\sum_i \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2$$

Physics-constrained model: exact solution of the balance ($T(0) = T_0$ by construction):

$$T(t) = T_{ss} + (T_0 - T_{ss})\,e^{-\kappa t/C}, \quad \kappa = h - G\eta_{ref}\beta$$

It learns only $\theta = (h_0, k, C)$ plus a bounded residual Δh:

$$\mathscr{L} = \overline{(\hat{T}_{15} - T_{15})^2} + \lambda_p \sum_j \left(\frac{\theta_j - \theta_j^0}{\theta_j^0}\right)^2 + \lambda_r \overline{\Delta h^2}$$

MPC step over a 19 × 19 pitch–yaw grid, 15-min horizon:

$$(\phi, \psi)^* = \arg\max_{\phi,\psi}\; G(\phi, \psi)\,\eta\left(\hat{T}_{15}(\phi, \psi)\right)$$

Synthetic Data and Surrogate

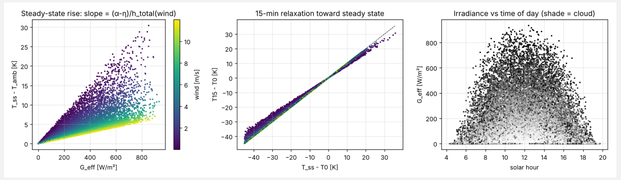


Figure 1. Training data obey the energy balance. Left: steady-state rise vs. irradiance; wind lowers the slope. Center: 15-min relaxation toward steady state. Right: irradiance vs. time of day (darker = clearer sky).

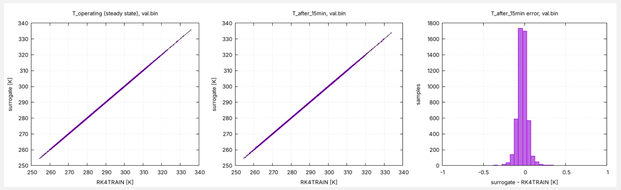


Figure 2. Surrogate vs. RK4 generator on held-out synthetic states. RMSE: 0.045 K (steady state), 0.061 K (15 min).

Key Finding: Temperature Cannot Move the Pose

Differentiating $P = G\,\eta(T(G))$ through the energy balance gives $dP/dG > 0$ for all

$$G < G^* = \frac{h\,\eta(T)}{\alpha\,\eta_{ref}\,\beta}.$$

- **Smallest G^* over 300 sweep conditions: 4,013 W/m^2.**
- **Largest available: 916 (model), 1,485 W/m^2 (Sandia).**
- **Max-power pose = max-irradiance pose in 100% of conditions.**

So the MPC is a sun tracker; the surrogate's useful output is the **15-min temperature forecast**.

Example Days (Sandia, held out)

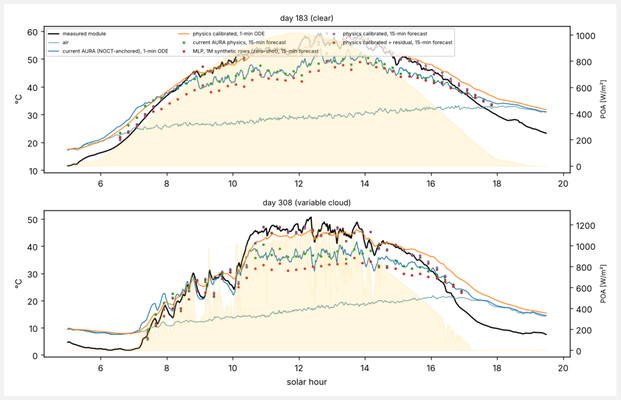


Figure 3. Measured module temperature (black) vs. models on a clear and a cloudy day. The calibrated model (orange) follows the measurement; the zero-shot MLP (red) runs cold.

Real-Data Validation: 15-min-Ahead Panel Temperature

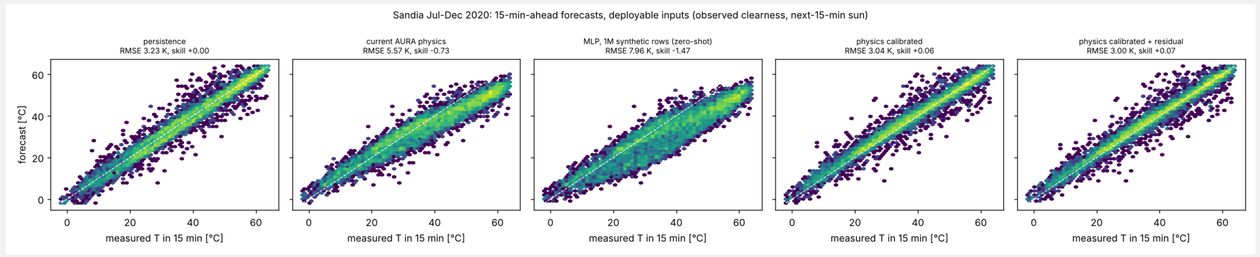


Figure 4. Forecast vs. measured temperature 15 min ahead (Sandia, Albuquerque; Driesse et al., 2023). Fit Jan–Jun 2020, scored Jul–Dec, deployable weather inputs. Color = point density. Skill = 1 − RMSE/RMSE$_{persistence}$.

RMSE [K]	perfect weather	deployable weather
Persistence	3.20	3.23
Current physics (NOCT)	5.12	5.57
MLP, zero-shot	7.51	7.96
Physics, calibrated	**2.21**	3.04
+ residual	**2.02**	**3.00**

- **The generator's wind slope is too steep: Sandia calibrates k = 4.12 vs. McAdams 6.78 W m^{-2}K^{-1}/(m/s).**
- **Calibrated physics beats persistence after only 100 real samples (perfect weather).**
- **A Faiman model fit to the same data gives 3.96 K (steady state; Faiman, 2008).**

Closed-Loop Sandbox

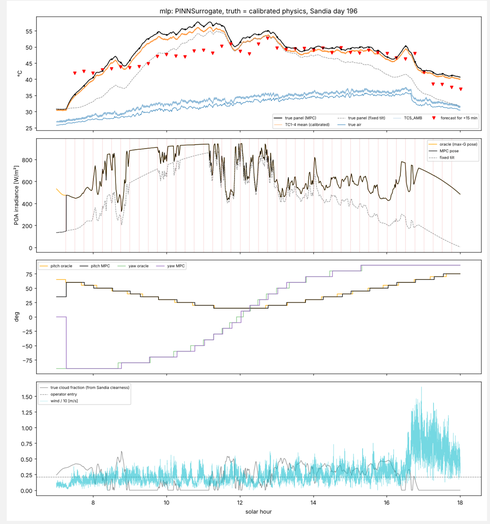


Figure 5. One simulated day (Sandia day 196). Top to bottom: panel temperature with +15-min forecasts (red); plane-of-array irradiance for MPC, oracle and fixed tilt; pitch and yaw; cloud and wind.

- **Simulates our DAQ: Pi Pico, 5 thermocouples, weather meter (cf. Asadi et al., 2016), with sensor errors and dropped packets.**
- **MPC / oracle energy: 0.987; fixed tilt / oracle: 0.678.**
- **Fixed-tilt gap is overstated (beam-only irradiance, no diffuse light).**

Conclusions

1. **The surrogate matches its generator to 0.06 K and retrains in about 2 min on a laptop CPU.**
2. **Under a power objective, the best pose is always the max-irradiance pose.**
3. **On real data, the zero-shot MLP is worse than persistence; calibrating 3 physical constants fixes it (2.21 K).**
4. **The MPC tracks the oracle closely in closed loop (98.7% of oracle energy).**

Next Steps

- **Collect thermocouple data on our own panel in Clarksville, TN.**
- **Refit k and C on it; regenerate data and retrain.**
- **Model pose-dependent convection (tilt vs. wind direction): the one way wind could move the pose. Add diffuse light.**

References

Asadi, A. A., et al. (2016). A data acquisition system based on Raspberry Pi: Design, construction and evaluation. *IEEE IEMCON*, 1–6.

Driesse, A., Theristis, M., & Stein, J. S. (2023). *PV module operating temperature: Data and resources*. Sandia National Laboratories. https://doi.org/10.21948/2204673

Faiman, D. (2008). Assessing the outdoor operating temperature of photovoltaic modules. *Progress in Photovoltaics, 16*(4), 307–315. https://doi.org/10.1002/pip.813

Laudani, A., Lozito, G. M., & Riganti Fulginei, F. (2021). Irradiance sensing through PV devices: A sensitivity analysis. *Sensors, 21*(13), 4264. https://doi.org/10.3390/s21134264

Nghiem, T. X., et al. (2023). Physics-informed machine learning for modeling and control of dynamical systems. *American Control Conference*, 3735–3750. https://doi.org/10.23919/ACC55779.2023.10155901

Sze, S. M., & Lee, M.-K. (2012). *Semiconductor devices: Physics and technology* (3rd ed.). Wiley.

Yadav, A., et al. (2023). The effect of temperature on photovoltaic power generation. *ICCIKE*, 45–49.

Acknowledgements

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